HERIOT WATT UNIVERSITY

School of Mathematical and Computer Sciences

Actuarial Mathematics and Statistics

F78PB

Probability and Statistics B

IN-PERSON EXAMINATION

April Examination and Assessment Diet 2024-2025

Duration: 2 Hours

Answer **ALL** 6 questions

Resources Included within Exam Paper: AMS formula sheet, & Correlation and Regression formula sheet
Additional Resources to be Supplied to Candidates: Blue Statistical Tables & Yellow Statistical Tables
Calculator Permitted: Approved calculators are permitted

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

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Some Continuous Distributions

Name	Notation	p.d.f.	Range	m.g.f.	Mean	Variance
Uniform	$\mathrm{U}(a,b)$	$\frac{1}{b-a}$	$a \le x \le b$	$\frac{e^{bt}-e^{at}}{(b-a)t}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	$\mathrm{N}(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	$-\infty < x < \infty$	$e^{\mu t+\frac{1}{2}\sigma^2 t^2}$	μ	σ^2
Log Normal	$\mathrm{LogN}(\mu,\sigma^2)$	$\frac{1}{x\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(\ln x-\mu)^2}{2\sigma^2}\right\}$	$0 < x < \infty$	undefined for $t > 0$	$e^{\mu+\frac{1}{2}\sigma^2}$	$(e^{\sigma^2}-1)(e^{2\mu+\sigma^2})$
Gamma	$\mathrm{Gamma}(\alpha,\lambda)$	$\frac{\lambda^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\lambda x}$	$0<x<\infty$, $\alpha,\lambda>0$	$\left(\frac{\lambda}{\lambda-t}\right)^\alpha$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
Exponential	$\mathrm{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$0<x<\infty$, $\lambda>0$	$\frac{\lambda}{\lambda-t}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Chi-square	χ^2_ν	$\frac{1}{2^{\nu/2}\Gamma(\nu/2)}x^{\nu/2-1}e^{-x/2}$	$0<x<\infty$, $\nu>0$ integer	$(1-2t)^{-\nu/2}$	ν	2ν
Beta	$\mathrm{Beta}(\alpha,\beta)$	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}$	$0<x<1$, $\alpha,\beta>0$	*omitted	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Cauchy	$\mathrm{Cauchy}(\mu,\sigma)$	$\frac{\sigma}{\pi(\sigma^2+(x-\mu)^2)}$	$\sigma > 0$	does not exist		
Pareto	$\mathrm{Pareto}(k,\theta)$	$\frac{k}{\theta}\left(1+\frac{x}{\theta}\right)^{-(k+1)}$	$x>0$, $\theta,k>0$	does not exist	$\frac{\theta}{k-1}$	$\frac{\theta^2 k}{(k-2)(k-1)^2}$
Weibull	$\mathrm{Weibull}(\nu,\lambda)$	$\nu\lambda x^{\nu-1}\exp\left(-\lambda x^\nu\right)$	$x>0$, $\nu,\lambda>0$	*omitted	$\lambda^{-1/\nu}\Gamma\left(1+\frac{1}{\nu}\right)$	*omitted
Multivariate Normal	$\mathrm{N}_p(\boldsymbol{\mu},\boldsymbol{\Sigma})$	$\frac{\exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}}{(2\pi)^{p/2}\det(\boldsymbol{\Sigma})^{1/2}}$	$\mathbf{x}\in\mathbb{R}^p$	$e^{\boldsymbol{\mu}^{\mathrm{T}}\mathbf{t}+\frac{1}{2}\mathbf{t}^{\mathrm{T}}\boldsymbol{\Sigma}\mathbf{t}}$	$\boldsymbol{\mu}$	covariance matrix $\boldsymbol{\Sigma}$

Some Discrete Distributions

Name	Notation	p.m.f.	Range	m.g.f.	Mean	Variance
Discrete Uniform	$\mathrm{DU}(a,b)$	$\frac{1}{b-a+1}$	$x = a,a+1,\ldots,b-1,b$ a,b integers	$\frac{e^{at}-e^{(b+1)t}}{(b-a+1)(1-e^t)}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2-1}{12}$
Binomial	$\mathrm{Bin}(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	$x = 0,1,2,\ldots,n$ $0<p<1$	$[(1-p)+pe^t]^n$	np	$np(1-p)$
Poisson	$\mathrm{Poi}(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$x = 0,1,2,\ldots$ $\lambda>0$ real	$e^{\lambda(e^t-1)}$	λ	λ
Hyper-geometric	$\mathrm{HypGeo}(n,K,N)$	$\frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$	$x = 0,1,2,\ldots,n$ N,K,n integers	*omitted	$\frac{nK}{N}$	$\frac{nK(N-K)(N-n)}{(N)^2(N-1)}$
Geometric	$\mathrm{Geo}(p)$	$p(1-p)^{x-1}$	$x = 1,2,\ldots$ $0<p<1$	$p[e^{-t}-(1-p)]^{-1}$	$1/p$	$(1-p)/p^2$
Negative Binomial	$\mathrm{NegBin}(k,p)$	$\binom{x-1}{k-1}p^k(1-p)^{x-k}$	$x = k,k+1,\ldots$ $0<p<1$	$p^k[e^{-t}-(1-p)]^{-k}$	k/p	$k(1-p)/p^2$

*omitted due to lack of space

Multinomial: multivariate extension of binomial

Probability mass function (p.m.f) is given by

$$f(x_1,x_2,\ldots,x_k)=\frac{n!}{x_1!x_2!\ldots x_k!}p_1^{x_1}p_2^{x_2}\ldots p_k^{x_k},\qquad 0<p_i<1,\forall i=1,2,\ldots,k$$

where $p_1+p_2+\cdots+p_k=1$ and $x_1+x_2+\cdots+x_k=n$.

The m.g.f. is given by

$$[p_1e^{t_1}+p_2e^{t_2}+\cdots+p_ke^{t_k}]^n.$$

Moments: $\mathbb{E}(X_i)=np_i$, $\mathbb{V}\mathrm{ar}(X_i)=np_i(1-p_i)$, $\mathbb{C}ov(X_i,X_j)=-np_ip_j$, $i,j=1,2,\ldots,n$.

Correlation and Regression

Basic sums:

$$S_{xy} = \sum x_i y_i - \frac{1}{n}(\sum x_i)(\sum y_i)$$

$$S_{xx} = \sum x_i^2 - \frac{1}{n}(\sum x_i)^2 \quad \text{and} \quad S_{yy} = \sum y_i^2 - \frac{1}{n}(\sum y_i)^2$$

Correlation coefficient:

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} \quad \text{and} \quad t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2} \quad \text{under } H_0 : \rho = 0$$

Under the assumption that (x,y) has the bivariate normal distribution, we can test $H_0 : \rho = \rho_0$ against $H_1 : \rho \neq \rho_0$ using the approximation:

$$W = \frac{1}{2}\log\frac{1+r}{1-r} \sim N\left(\frac{1}{2}\log\frac{1+\rho}{1-\rho}, \frac{1}{n-3}\right)$$

Regression parameter estimates:

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} \quad \text{and} \quad \hat{\alpha} = \frac{1}{n}\{\sum y_i - \hat{\beta}\sum x_i\}$$

$$\hat{\sigma}^2 = \frac{1}{n-2}\{S_{yy} - \frac{S_{xy}^2}{S_{xx}}\}$$

Residuals and fitted values:

Fitted values $\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$ and Residuals $\hat{e}_i = y_i - \hat{y}_i$

Slope coefficient:

$$t = \frac{\hat{\beta} - \beta}{se(\hat{\beta})} \sim t_{n-2} \quad \text{where} \quad se(\hat{\beta}) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}$$

Prediction standard errors:

For <u>expected</u> *Y* at $X = x_0$: $\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$ and for <u>individual</u> *Y*: $\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} + 1}$

1. Let $Z \sim \mathrm{N}(0,1)$ be a standard normal random variable, then the moment generating function, $M_Z(\theta)$, of Z is given by

$$M_Z(\theta) = \exp\left(\frac{1}{2}\theta^2\right).$$

(a) Let $X = \mu + \sigma Z \sim \mathrm{N}(\mu, \sigma^2)$. Use the moment generating function for Z to show that the moment generating function $M_X(\theta)$ for X is given by

$$M_X(\theta) = e^{\mu\theta + \frac{1}{2}\sigma^2\theta^2}.$$

[3 marks]

(b) For $X \sim \mathrm{N}(1,4)$ (with $\mu = 1$ and $\sigma = 2$) use the series expansion of the moment generating function, or otherwise, to calculate the third moment $\mathbb{E}[X^3]$.

[5 marks]

(c) It is known that for $Z \sim \mathrm{N}(0,1)$ the moment generating function, $M_{Z^2}(\theta)$, of Z^2 is given by

$$M_{Z^2}(\theta) = \frac{1}{\sqrt{1-2\theta}}$$

Let $Z_1, \ldots, Z_n$ be a sequence of independent standard normal random variables, and define

$$S = Z_1^2 + Z_2^2 + \ldots + Z_n^2.$$

Determine the moment generating function of S. Hence, or otherwise, state the distribution of S.

[3 marks]

[Total 11 marks]

2. An engineer wants to estimates the maximum stress β that a construction material can withstand before failure. The stress measurements $X_1, X_2, \ldots X_n$ follow a probability distribution with mean and variance given by:

$$\mathbb{E}(X) = \frac{3\beta}{4} \text{ and } \mathbb{V}\text{ar}(X) = \frac{3\beta^2}{80}$$

Consider an estimator $\hat{\beta}_1$ for β:

$$\hat{\beta}_1 = \frac{4\sum_{i=1}^{n} X_i}{3n} = \frac{4\bar{X}}{3}$$

(a) i. Is $\hat{\beta}_1$ an unbiased estimator of β? Justify your answer.

[2 marks]

ii. Calculate the mean square error (MSE) of $\hat{\beta}_1$.

[2 marks]

(b) Consider another estimator of the form $\hat{\beta}_2 = \gamma\bar{X}$ where $\gamma > 0$ is a given constant.

i. Calculate the MSE of $\hat{\beta}_2$ in terms of the constant γ.

[2 marks]

ii. Find the value of γ which makes $\hat{\beta}_2$ the most efficient estimator of β.

[4 marks]

[Total 10 marks]

3. A scientist is studying the success rate of pollination for a rare species of flower in a particular region. Specifically, the researcher is focusing on the proportion of flowers that get pollinated during a specific time period. The pollination success rate for each flower, denoted as $X_1, X_2, \ldots, X_n$ follows a probability density function (pdf) defined by:

$$f(x \mid \theta) = \frac{1}{\theta} x^{(1-\theta)/\theta}, \quad 0 < x < 1, 0 < \theta < \infty$$

where θ is an unknown parameter and has to be estimated from the data.

(a) Derive the maximum likelihood estimator for θ.

[5 marks]

(b) Derive the method of moment estimator for θ.

[4 marks]

(c) Suppose three independent random samples obtained from the distribution $f_X(x)$ are: $0.11, 0.22,$ and 0.33. Which of the two estimators in **3(a)** and **3(b)** provide the smallest estimate of θ? Justify your answer

[2 marks]

[Total 11 marks]

4. (a) A company tests the effect of a new training programme on employee performance. The performance scores before and after the training programme are recorded for 12 employees. The following is a summary of the scores:

$$n = 12, \bar{D} = 7, s_D^2 = 18$$

where n, $\bar{D}$, and s_D^2 represent the sample size, mean, and variance, respectively, for the difference in performance scores after and before the training programme. Assume that the scores are normally distributed.

i. Explain why

$$\frac{\bar{D} - \mu_D}{\sqrt{S_D^2/n}}$$

is a pivotal quantity for $\bar{D}$, the mean difference in paired observations.

[3 marks]

ii. For the above data, calculate the 95% confidence interval for the mean difference in performance scores after and before the training programme. Does the calculated 95% confidence interval suggest that the training programme has a significant effect on employee performance?

[3 marks]

(b) A company measures the strength of two different types of materials used in construction. The following data represent the sample variances of the strength measurements for the two materials based on independent samples:

$$\text{Material X: } n_X = 9, s_X^2 = 25$$
$$\text{Material Y: } n_Y = 8, s_Y^2 = 40$$

where n and s^2 represent the sample size and variance, respectively. Using the F-distribution, construct a 95% confidence interval for the ratio of the population variances of the two materials, and determine if there is a significant difference in the variability of the strength measurements between the two materials.

[4 marks]

[Total 10 marks]

5. A researcher wants to compare the average test scores of two groups of students who were taught using different teaching methods. Group A used a traditional teaching method, and Group B used an innovative method. The researcher collects the following data:

$$\begin{aligned} &\text{Group A:} \quad n_1 = 10, \bar{x}_1 = 75, s_1 = 7 \\ &\text{Group B:} \quad n_2 = 12, \bar{x}_2 = 80, s_2 = 6 \end{aligned}$$

where n, $\bar{x}$, and s represent the sample size, mean, and standard deviation, respectively. Conduct a suitable hypothesis test to determine if the innovative teaching method (Group B) results in a significantly higher average test score than the traditional method, by doing the following steps:

(a) State the three assumptions required to perform the hypothesis test.

[3 marks]

(b) State the hypotheses clearly and compute the test statistics.

[5 marks]

(c) Calculate the p-value corresponding to the test statistic and draw a conclusion about the research hypothesis.

[2 marks]

[Total 10 marks]

6. The marketing department of a company is determining the size of the budget to spend on advertising on different media options. They have data (x_i, y_i) from 15 companies. Here x_i denotes the amount spent on newspapers with units $10,000$\$, and y_i denotes the sales amount with units 1000\$. For this data

$$\sum x_i = 58, \qquad \sum x_i^2 = 350,$$
$$\sum y_i = 210, \qquad \sum y_i^2 = 3300,$$
$$\sum x_i y_i = 830.$$

(a) Calculate the sample correlation coefficient for the amount spent on newspapers, x_i, versus the sales figures, y_i. Interpret what this value tells you.

[4 marks]

(b) Perform a statistical test to determine if the correlation coefficient is significantly different from 0 at a 5% level of significance.

[4 marks]

[Total 8 marks]

END OF PAPER